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Inventor(s)

Leck; Chris James et al.

TRACTION BATTERY BUSBAR ASSEMBLY LOCKING BAR

Abstract

A traction battery pack assembly includes a busbar assembly having at a frame and at least one busbar. The frame includes a plurality of fingers. A locking bar has a plurality of apertures that are each configured to receive a portion of one or more of the fingers to constrain the busbar assembly.

Inventors: Leck; Chris James (Enfield, GB), Macwillson; Hamish (Colchester, GB), Thomas; Ronnie (South Woodham Ferrers, GB)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to using a locking bar to constrain movement of a busbar assembly of a traction battery pack, particularly vibrations of the busbar assembly.

BACKGROUND

[0002] Electrified vehicles differ from conventional motor vehicles because electrified vehicles can be selectively driven by one or more electric machines that are powered by a traction battery pack. The electric machines can propel the electrified vehicles instead of, or in combination with, an internal combustion engine. The traction battery pack is discharged when powering the one or more electric machines and other loads of the electrified vehicle.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a traction battery pack assembly, including: a busbar assembly having at a frame and at least one busbar, the frame including a plurality of fingers; and a locking bar have a plurality of apertures that are each configured to receive a portion of one or more of the fingers to constrain the busbar assembly.

[0004] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the busbar assembly is pivotably coupled to a cover that spans a cell stack.

[0005] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the locking bar is a first locking bar and the plurality of apertures are a first plurality of apertures, the busbar assembly is a first busbar assembly pivotably coupled to a first side of the cover, the frame is a first frame, the plurality of fingers is a plurality of first fingers, and further including a second busbar assembly and a second locking bar, the second busbar assembly having a second frame with a second plurality of fingers and at least one second busbar, the second locking bar having a second plurality of apertures, the second plurality of apertures configured to receive a portion of one or more of the second plurality of fingers to constrain the second busbar assembly.

[0006] In some aspects, the techniques described herein relate to a traction battery pack assembly, further including a cell stack, the first busbar assembly and first locking bar disposed along a first side of the cell stack, the second busbar assembly and the second locking bar disposed along an opposite, second side of the cell stack.

[0007] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the first busbar assembly is pivotably coupled to a first side of a cover that spans the cell stack, and the second busbar assembly is pivotably coupled to an opposite, second side of the cover.

[0008] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein each of the apertures within the plurality of apertures receive one of the fingers within the plurality of fingers.

[0009] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the plurality of fingers are tapered.

[0010] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the plurality of apertures are a plurality of pockets, the locking bar disposed circumferentially completely about the portion of one or more of the fingers within the plurality of fingers to constrain the busbar assembly.

[0011] In some aspects, the techniques described herein relate to a traction battery pack assembly, further including a cell stack having a plurality of battery cells disposed along an axis, the plurality of battery cells positioned between compressive structures that are at opposing axial ends of the cell stack, the busbar assembly and locking bar disposed along a side of the cell stack, the locking bar clipped to the compressive structures.

[0012] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the compressive structures are cross-members.

[0013] In some aspects, the techniques described herein relate to a traction battery pack assembly,

further including a cell stack having a plurality of battery cells with tab terminals, wherein the frame includes a plurality of slots the tab terminals received within the plurality of slots.

[0014] In some aspects, the techniques described herein relate to a method of assembling a battery pack, including: securing a locking bar relative to a cell stack; moving a busbar assembly into an installed position relative to the cell stack, the busbar assembly including a plurality of busbars mounted to a frame; and during the moving, inserting a plurality of fingers of the frame into a plurality of apertures of the locking bar.

[0015] In some aspects, the techniques described herein relate to a method, further including inhibiting vibration of the plurality of fingers using the locking bar.

[0016] In some aspects, the techniques described herein relate to a method, further including constraining movement of the plurality of fingers using the locking bar.

[0017] In some aspects, the techniques described herein relate to a method, further including clipping the locking bar to at least one compressive structure when securing the locking bar.

[0018] In some aspects, the techniques described herein relate to a method, further including securing a plurality of tab terminals of the cell stack to the plurality of busbars.

[0019] In some aspects, the techniques described herein relate to a method, wherein the securing including welding the plurality of tab terminals to at least one of the busbars within the plurality of busbars.

[0020] In some aspects, the techniques described herein relate to a method, wherein the busbar assembly is pivotably secured to a cover that spans a cell stack when the plurality of fingers are received within the plurality of apertures of the locking bar.

[0021] In some aspects, the techniques described herein relate to a method, wherein the busbar assembly is a first busbar assembly disposed along a first side of the cell stack, and further including disposing a second busbar assembly along an opposite, second side of the cell stack, the second busbar pivotably secured to the cover.

[0022] The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0023] The various features and advantages of the disclosed examples will become apparent to those skilled in the art from the detailed description. The figures that accompany the detailed description can be briefly described as follows:

[0024] FIG. 1 illustrates a side view of an electrified vehicle.

[0025] FIG. 2 illustrates an expanded, perspective view of a battery pack from the electrified vehicle of FIG. 2 according to an exemplary embodiment of the present disclosure.

[0026] FIG. 3 illustrates a battery cell from the battery pack of FIG. 2.

[0027] FIG. 4 illustrates a locking bar from the battery pack of FIG. 2.

[0028] FIG. 5 illustrates a portion of a busbar module in an installed position where fingers of the busbar module are constrained by the locking bar of FIG. 4.

[0029] FIG. 6 illustrates a section view taken at line 6-6 in FIG. 5.

DETAILED DESCRIPTION

[0030] This disclosure details example locking bars that can be used to constrain busbar modules within a battery pack and inhibit vibration. The locking bars can help support the busbar modules when tab terminals are secured to busbars of the busbar modules.

[0031] With reference to FIG. 1, an electrified vehicle 10 includes a battery pack 14, an electric

machine **18**, and wheels **22**. The battery pack **14** powers an electric machine **18**, which can convert electrical power to mechanical power to drive the wheels **22**. The battery pack **14** is thus a traction battery pack.

[0032] The battery pack **14** is, in the exemplary embodiment, secured to an underbody **26** of the electrified vehicle **10**. The battery pack **14** could be located elsewhere on the electrified vehicle **10** in other examples.

[0033] The electrified vehicle **10** is an all-electric vehicle. In other examples, the electrified vehicle **10** is a hybrid electric vehicle, which selectively drives wheels using torque provided by an internal combustion engine instead of, or in addition to, an electric machine. Generally, the electrified vehicle **10** could be any type of vehicle having a battery pack.

[0034] With reference now to FIGS. **2** and **3**, the battery pack **14** includes a plurality of cell stacks **30** held within an enclosure assembly **34**. In the exemplary embodiment, the enclosure assembly **34** includes an enclosure cover **38** and an enclosure tray **42**. The enclosure cover **38** can be secured to the enclosure tray **42** to provide an interior area **44** that houses the cell stacks **30**. The enclosure cover **38** can be secured to the enclosure tray **42** using mechanical fasteners (not shown), for example.

[0035] Each of the cell stacks **30** includes, among other things, a plurality of battery cells **50** (or simply “cells”) stacked side-by-side relative to each along a respective cell stack axis A. In this example, the axes A of the cell stacks **30** extend longitudinally from a forward end to a rearward end of the vehicle **10**.

[0036] The battery cells **50** store and supply electrical power. Although a specific number of the cell stacks **30** and cells **50** are illustrated in the various figures of this disclosure, the battery pack **14** could include any number of the cell stacks **30** each having any number of individual cells **50**.

[0037] In an embodiment, the battery cells **50** are lithium-ion pouch cells. However, battery cells having other geometries (cylindrical, prismatic, etc.), other chemistries (nickel metal hydride, lead acid, etc.), or both could be alternatively utilized within the scope of this disclosure.

[0038] With reference now to FIGS. **4-6** and continuing reference to FIGS. **1-3**, the cell stacks **30** are each sandwiched along their axis A between compressive structures **52**, which are disposed at opposing axial ends of the cell stacks **30**. In this example, the compressive structures **52** are cross-members within the enclosure tray **42**. In another example, the compressive structures **52** are part of the enclosure assembly **34**, such as the enclosure cover **38** or the enclosure tray **42**. In still other examples, the compressive structures **52** are endplates. The compressive structures **52** are compressive as the compressive structures **52** maintain the cell stack **30** in a compressed state along the respective axis A.

[0039] Each of the example battery cells **50** includes a pair of tab terminals **54** extending from a case **56**. Within a given one of the cell stacks **30**, the individual battery cells **50** can be electrically connected together. To provide these electrical connections, the tab terminals **54** of the battery cells **50** can be connected to the tab terminals **54** of other battery cells **50**.

[0040] In this example, an interconnector assembly **62** is associated with each of the cell stacks **30**. The interconnector assemblies **62** each include a cover **66** and a pair of busbar assemblies **70**. One of the busbar assemblies **70** is pivotably coupled to each side of the cover **66**. When the interconnector assembly **62** is installed, the cover **66** of the interconnector assembly **62** spans the respective cell stack **30**, with the busbar assemblies **70** disposed along opposite sides of the respective cell stack **30**.

[0041] The busbar assemblies **70** each include a frame **74** and a plurality of busbars **78** mounted to the frame **74**. In this example, the busbars **78** are mounted to the frame **74** using heat stakes.

[0042] The frame **74** includes a plurality of fingers **82**. Slots **86** are positioned between the fingers **82**. When the interconnector assembly **62** is moved to an installed position, the tab terminals **54** are received within one of the slots **86**. The tab terminals **54** are then folded over and secured to one of the busbars **78**. In this example, the tab terminals **54** are welded to the busbars **78** after being folded

over.

[0043] The battery pack **14** includes locking bars **90** that are disposed along each side of the cell stacks **30**. The locking bars **90** include a plurality of apertures **94**. In this example, the apertures **94** are pockets that open vertically upward, and that do not open to a side of the locking bars **90**. In another example, the apertures **94** could be slots that open to one or more sides of the locking bars **90**. Vertical and horizontal, for purposes of this disclosure, is with reference to ground and a general orientation of the vehicle **10** and battery pack **14** during operation.

[0044] As the busbar assemblies **70** are moved to an installed position where the tab terminals **54** are received within the slots **86** of the frame **74**, a portion of each of the fingers **82** is inserted into one of the apertures **94**. When the portion of the fingers **82** is received with each of the apertures **94**, movement of the fingers **82** is constrained. The fingers **82** are tapered to facilitate insertion into the apertures **94**.

[0045] In this example, the locking bars **90** constrain at least horizontal movements and vibrations of the fingers **82**. Each of the apertures **94** receives a portion of one of the fingers **82**—here a tip of the fingers **82**. In another example, the apertures **94** could receive portions of more than one of the fingers **82**.

[0046] When the fingers **82** are received within the apertures **94**, the locking bars **90** are disposed circumferentially completely about the portion of one or more of the fingers **82** to constrain the busbar assembly **70**.

[0047] Constraining movement of the fingers **82** can help to inhibit vibration of the fingers **82** when, for example, the vehicle **10** is driven. Constraining movement of the fingers **82** can also help brace the fingers **82** when the tab terminals **54** are welded to the busbars **78**.

[0048] The locking bars **90**, in this example, clip into the compressive structures **52** to secure the locking bars **90**. In this example, protrusions **96** at opposite ends of the locking bars **90** are received within apertures **98** of the compressive structures **52** to clip the locking bars **90** into the compressive structures **52**. The locking bars **90** could instead or additionally be mechanically fastened to the enclosure tray **42** using threaded fasteners, such as bolts.

[0049] The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from the essence of this disclosure. Thus, the scope of protection given to this disclosure can only be determined by studying the following claims.

Claims

1. A traction battery pack assembly, comprising: a busbar assembly having at a frame and at least one busbar, the frame including a plurality of fingers; and a locking bar have a plurality of apertures that are each configured to receive a portion of one or more of the fingers to constrain the busbar assembly.
2. The traction battery pack assembly of claim 1, wherein the busbar assembly is pivotably coupled to a cover that spans a cell stack.
3. The traction battery pack assembly of claim 2, wherein the locking bar is a first locking bar and the plurality of apertures are a first plurality of apertures, the busbar assembly is a first busbar assembly pivotably coupled to a first side of the cover, the frame is a first frame, the plurality of fingers is a plurality of first fingers, and further comprising a second busbar assembly and a second locking bar, the second busbar assembly having a second frame with a second plurality of fingers and at least one second busbar, the second locking bar having a second plurality of apertures, the second plurality of apertures configured to receive a portion of one or more of the second plurality of fingers to constrain the second busbar assembly.
4. The traction battery pack assembly of claim 3, further comprising a cell stack, the first busbar assembly and first locking bar disposed along a first side of the cell stack, the second busbar

- assembly and the second locking bar disposed along an opposite, second side of the cell stack.
- 5.** The traction battery pack assembly of claim 3, wherein the first busbar assembly is pivotably coupled to a first side of a cover that spans the cell stack, and the second busbar assembly is pivotably coupled to an opposite, second side of the cover.
- 6.** The traction battery pack assembly of claim 1, wherein each of the apertures within the plurality of apertures receive one of the fingers within the plurality of fingers.
- 7.** The traction battery pack assembly of claim 1, wherein the plurality of fingers are tapered.
- 8.** The traction battery pack assembly of claim 1, wherein the plurality of apertures are a plurality of pockets, the locking bar disposed circumferentially completely about the portion of one or more of the fingers within the plurality of fingers to constrain the busbar assembly.
- 9.** The traction battery pack assembly of claim 1, further comprising a cell stack having a plurality of battery cells disposed along an axis, the plurality of battery cells positioned between compressive structures that are at opposing axial ends of the cell stack, the busbar assembly and locking bar disposed along a side of the cell stack, the locking bar clipped to the compressive structures.
- 10.** The traction battery pack assembly of claim 9, wherein the compressive structures are cross-members.
- 11.** The traction battery pack assembly of claim 1, further comprising a cell stack having a plurality of battery cells with tab terminals, wherein the frame includes a plurality of slots the tab terminals received within the plurality of slots.
- 12.** A method of assembling a battery pack, comprising: securing a locking bar relative to a cell stack; moving a busbar assembly into an installed position relative to the cell stack, the busbar assembly including a plurality of busbars mounted to a frame; and during the moving, inserting a plurality of fingers of the frame into a plurality of apertures of the locking bar.
- 13.** The method of claim 12, further comprising inhibiting vibration of the plurality of fingers using the locking bar.
- 14.** The method of claim 12, further comprising constraining movement of the plurality of fingers using the locking bar.
- 15.** The method of claim 12, further comprising clipping the locking bar to at least one compressive structure when securing the locking bar.
- 16.** The method of claim 12, further comprising securing a plurality of tab terminals of the cell stack to the plurality of busbars.
- 17.** The method of claim 16, wherein the securing comprising welding the plurality of tab terminals to at least one of the busbars within the plurality of busbars.
- 18.** The method of claim 12, wherein the busbar assembly is pivotably secured to a cover that spans a cell stack when the plurality of fingers are received within the plurality of apertures of the locking bar.
- 19.** The method of claim 18, wherein the busbar assembly is a first busbar assembly disposed along a first side of the cell stack, and further comprising disposing a second busbar assembly along an opposite, second side of the cell stack, the second busbar pivotably secured to the cover.
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